

NCT07562841: Au-TMP and Radiotherapy for Advanced Melanoma With Anti-PD-1 Therapy

Gender: ALL

Age: 18 Years - MISSING

Criteria:

Inclusion Criteria:

1. Age: Age $\geq$ 18 years.
2. Eastern Cooperative Oncology Group (ECOG) performance status of 0 or 1.
3. Histologically confirmed unresectable Stage III or Stage IV melanoma without prior systemic therapy. Prior adjuvant or neoadjuvant therapy is permitted, provided it was completed at least 3 weeks before enrollment and all related adverse events (AEs) have resolved to baseline or NCI CTCAE v5.0 Grade $\leq$ 1.
4. Presence of at least one measurable lesion according to RECIST v1.1 criteria.
5. At least one lesion suitable for intratumoral injection and radiotherapy (located in the skin, subcutaneous tissue, superficial lymph nodes, or visceral lesions assessed as safe for access) that has not received prior radiotherapy (unless documented progression has occurred).
6. Adequate hematologic and organ function within 7 days prior to the first dose, including:

Hematology: Absolute Neutrophil Count (ANC) $\geq$ 1.5 $\times 10^9$ /L; Platelet count (PLT) $\geq$ 90 $\times 10^9$ /L; Hemoglobin (Hb) $\geq$ 90 g/L. (No G-CSF, platelet transfusion, or erythropoietin/RBC transfusion within 14 days prior to testing).

Renal Function: Serum creatinine (Cr) $\leq$ 1.5 Upper Limit of Normal (ULN), or calculated creatinine clearance (Ccr) $\geq$ 50 mL/min using the Cockcroft-Gault formula.

Hepatic Function: Total bilirubin (TBIL) $\leq$ 1.5 ULN ($\leq$ 3.0 ULN for patients with Gilbert's Syndrome or liver metastases); AST, ALT, and ALP $\leq$ 2.5 ULN ($\leq$ 5.0 ULN for patients with documented liver or bone metastases); Serum albumin $\geq$ 2.8 g/dL.

Coagulation: INR or Prothrombin Time (PT) and activated Partial Thromboplastin Time (aPTT) $\leq$ 1.5 ULN.

Cardiac: Left Ventricular Ejection Fraction (LVEF) $\geq$ 50%.

7. Anticipated survival time $\geq$ 16 weeks.
8. Agreement to use highly effective contraception methods during the trial and for 12 months after the last dose of treatment.
9. Voluntarily participate in the study, sign the Informed Consent Form (ICF), demonstrate good compliance, and be willing to cooperate with follow-up.

Exclusion Criteria:

1. Prior treatment with anti-PD-1, anti-PD-L1, or anti-PD-L2 agents.
2. Known hypersensitivity to recombinant humanized anti-PD-1 monoclonal antibodies or any of their components.
3. Active skin breakdown, infection, ulceration, necrosis, bleeding at the injection site, or a high risk of hollow organ perforation.
4. Known allergy or intolerance to Au-TMP active ingredients, excipients, or similar compounds.
5. Presence of known driver mutations (e.g., BRAF V600E/K, c-KIT, NRAS) for which targeted therapies are already approved and available as first-line treatment.
6. Ocular (uveal) or mucosal melanoma.
7. Receipt of other anti-tumor therapies (including corticosteroids or immunotherapy) or participation in other clinical trials within 4 weeks before treatment initiation; failure to recover from toxicities of prior therapies (except Grade 2 alopecia and Grade 1 neurotoxicity).
8. Pregnant or breastfeeding women.

9. Positive for HIV or HCV. Patients with positive HBsAg or HBcAb must have a negative HBV DNA test (quantitative detection $\leq$ 500 IU/mL).
10. History of active tuberculosis.
11. Active autoimmune disease requiring systemic treatment within the past 2 years (except physiological replacement therapy for thyroid, insulin, or adrenal/pituitary insufficiency).
12. Serious uncontrolled concurrent medical conditions, including uncontrolled diabetes, interstitial lung disease, NYHA Class III/IV heart failure, severe cardiac arrhythmias, or recent (within 6 months) myocardial infarction or cerebrovascular accidents.
13. Active Central Nervous System (CNS) or leptomeningeal metastases. Patients with treated brain metastases are eligible if they are stable (no progression via MRI) for at least 8 weeks post-treatment and 28 days prior to the first dose, and do not require immunosuppressive doses of corticosteroids ($\geq$ 10 mg/day prednisone equivalent) for at least 2 weeks.
14. Receipt of hematopoietic stimulants (e.g., G-CSF, EPO) within 2 weeks prior to treatment.
15. Receipt of live vaccines within 4 weeks prior to treatment.
16. Major surgery (excluding diagnostic procedures) within 4 weeks prior to treatment.
17. History of psychiatric disorders or persistent drug/substance abuse.
18. Other malignancies within the past 5 years, except for successfully treated localized cancers such as basal/squamous cell skin cancer or in situ carcinomas (cervix, breast, prostate).
19. Any other acute or chronic medical/psychiatric condition or laboratory abnormality that, in the investigator's opinion, increases research-related risk or interferes with the interpretation of study results.
20. Any condition that is not in the best interest of the participant.